

SIMATS ENGINEERING

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DATA INTERPRETATION

1. Standardization / Normalization

A feature "Age" ranges from 18–75 and another feature "Income" ranges from ₹20,000–₹20,00,000 in a dataset meant for a distance-based algorithm. Explain, with reasoning (not just definitions), why failing to standardize these variables would distort analysis, and calculate the z-score for Age = 45 if mean = 40 and SD = 12.

Solution

In a distance-based algorithm, the distance between observations is calculated using the numerical values of the features. Therefore, features with much larger numerical ranges can contribute much more to the calculated distance.

Age ranges only from 18 to 75, whereas Income ranges from ₹20,000 to ₹20,00,000. The numerical scale of Income is therefore much larger than that of Age. If the variables are used without standardization, changes in Income can dominate the distance calculation, while Age may contribute comparatively little.

As a result, the algorithm may treat two observations as similar or different mainly because of their income values rather than considering both variables fairly. Standardization places the variables on a comparable scale, allowing each feature to contribute according to its relative deviation from its mean.

Z-score Calculation

Formula:

$$z = (X - \mu) / \sigma$$

Given:

$$X = 45, \mu = 40, \sigma = 12$$

$$z = (45 - 40) / 12 = 5 / 12 = 0.4167$$

Therefore, the z-score is approximately 0.42. This means that an age of 45 is about 0.42 standard deviations above the mean age of 40.

Conclusion

Standardization is important for distance-based analysis because it prevents a feature with a large numerical scale, such as Income, from dominating a feature such as Age. The z-score for Age = 45 is approximately 0.42.

2. Distribution Shape from Summary Statistics Alone

A dataset has Mean = 45, Median = 45, Mode = 45, and skewness ≈ 0 , but kurtosis = 5.8 (leptokurtic). Explain what this combination tells you about the distribution's shape that skewness and central tendency alone would miss.

Solution

The mean, median, and mode are all equal to 45. Also, skewness is approximately zero. Together, these statistics indicate that the distribution is approximately symmetric around 45 and does not show a meaningful left or right tail imbalance.

However, skewness does not describe how concentrated the observations are around the centre or how heavy the tails are. This additional information is provided by kurtosis.

The kurtosis is 5.8, which is greater than the normal-distribution benchmark of 3 when using ordinary kurtosis. Therefore, the distribution is described as leptokurtic. It tends to have a sharper central concentration and heavier tails than a normal distribution, meaning that observations farther from the mean can occur more frequently than they would in a normal distribution.

Interpretation

Statistic	Interpretation
Mean = 45	Central average is 45
Median = 45	Middle value is 45
Mode = 45	Most frequent value is 45
Skewness ≈ 0	Distribution is approximately symmetric
Kurtosis = 5.8	Distribution is leptokurtic with heavier tails

Conclusion

Central tendency and skewness suggest a symmetric distribution centred at 45, but the high kurtosis reveals an important feature that these measures alone cannot show: the distribution is leptokurtic, with greater tail heaviness and a sharper central concentration.

3. Grouped Data — Mean and Variance

Class Interval: 0–10, 10–20, 20–30, 30–40, 40–50

Frequency: 5, 8, 15, 12, 10

Calculate the mean and variance using the step-deviation method. Show the assumed midpoints and working.

Solution

Step 1: Calculate the class midpoints.

Class Interval	Frequency (f)	Midpoint (x)	$d' = (x - A)/h$	fd'
0–10	5	5	-2	-10
10–20	8	15	-1	-8

20–30	15	25	0	0
30–40	12	35	1	12
40–50	10	45	2	20

Assume $A = 25$ as the assumed mean and $h = 10$ as the class width.

Total frequency:

$$\Sigma f = 5 + 8 + 15 + 12 + 10 = 50$$

From the table:

$$\Sigma fd' = -10 - 8 + 0 + 12 + 20 = 14$$

Mean using the step-deviation method:

$$\begin{aligned} \text{Mean} &= A + h(\Sigma fd' / \Sigma f) \\ &= 25 + 10(14 / 50) = 25 + 2.8 = 27.8 \end{aligned}$$

Therefore:

$$\text{Mean} = 27.8$$

For variance, calculate $(d')^2$ and fd'^2 :

d'	$(d')^2$	$f(d')^2$
-2	4	20
-1	1	8
0	0	0
1	1	12
2	4	40

Thus:

$$\Sigma f(d')^2 = 20 + 8 + 0 + 12 + 40 = 80$$

Variance formula using step-deviation:

$$\begin{aligned} \sigma^2 &= h^2[(\Sigma f(d')^2 / \Sigma f) - (\Sigma fd' / \Sigma f)^2] \\ &= 10^2[(80/50) - (14/50)^2] \\ &= 100[1.6 - 0.0784] = 152.16 \end{aligned}$$

Therefore, the grouped-data mean is 27.8 and the population variance is 152.16.

4. Comparing Two Distributions via Coefficient of Variation

Portfolio A: Mean return = 12%, SD = 3%

Portfolio B: Mean return = 18%, SD = 5%

An investor claims Portfolio B is "riskier in absolute terms but not necessarily worse per unit of return." Evaluate this claim quantitatively using Coefficient of Variation and explain what CV captures that SD alone does not.

Solution

The Coefficient of Variation (CV) compares the standard deviation with the mean. It therefore measures relative variability or risk per unit of average return.

$$\text{CV} = (\text{SD} / \text{Mean}) \times 100$$

Portfolio A

$$\text{CV_A} = (3 / 12) \times 100 = 25\%$$

Portfolio B

$$\text{CV_B} = (5 / 18) \times 100 \approx 27.78\%$$

Portfolio	Mean Return	CV
A	12%	25%
B	18%	27.78%

Portfolio B has a higher standard deviation (5% compared with 3%), so it is riskier in absolute terms. However, its CV is also higher: approximately 27.78% compared with 25% for Portfolio A.

Therefore, based on CV, Portfolio B is not better per unit of return. It has slightly more relative variability for each unit of expected return than Portfolio A.

SD alone measures absolute dispersion and does not account for the size of the mean return. CV accounts for both variability and the mean, making it useful when comparing investments with different average returns.

Conclusion

The investor's first statement is correct: Portfolio B is riskier in absolute terms. However, the second part is not supported by the CV values because Portfolio B has the higher relative risk (27.78% versus 25%).

5. Outlier's Effect on Variance

A dataset of 10 values has variance = 25. One value, originally 40 (close to the mean of 42), is mistakenly recorded as 400. Without recalculating exactly, explain directionally and reason about the magnitude of distortion to (a) the mean, (b) the variance — and why variance is affected disproportionately more than the mean.

Solution

The original value 40 is close to the mean of 42, so it contributes only a small deviation from the centre. When it is incorrectly changed to 400, it becomes an extremely large value relative to the other observations.

(a) Effect on the Mean

The original total of the 10 observations can be inferred from the mean of 42:

$$\text{Original total} = 10 \times 42 = 420$$

Replacing 40 by 400 increases the total by:

$$400 - 40 = 360$$

New total = $420 + 360 = 780$. Hence, the new mean becomes:

$$\text{New mean} = 780 / 10 = 78$$

Thus, the mean increases from 42 to 78. The mean is pulled upward substantially because the erroneous value is extremely large.

(b) Effect on the Variance

The variance is affected much more severely. Variance is based on squared deviations from the mean. Therefore, a very large deviation is squared, causing a disproportionately large contribution.

For example, relative to the original mean of 42, the original value 40 is only 2 units away:

$$(40 - 42)^2 = 4$$

The incorrectly recorded value 400 is 358 units above 42:

$$(400 - 42)^2 = 358^2 = 128,164$$

This illustrates why the variance can increase enormously. The exact new variance depends on the original observations and the variance convention being used; however, the direction and magnitude of the distortion are clear: the variance becomes dramatically larger.

Why Variance Is Affected More Than the Mean

The mean changes linearly with the value: replacing 40 with 400 changes the total by only 360, which changes the mean by 36 for 10 observations. Variance, in contrast, uses squared deviations. Squaring a

very large deviation magnifies its effect, so a single extreme error can cause variance to increase by a very large amount.

Conclusion

The incorrect recording of 400 pulls the mean upward and causes a much more severe increase in variance. The mean is affected directly by the magnitude of the erroneous value, while variance magnifies large deviations through squaring. Hence, variance is disproportionately sensitive to extreme values and data-entry errors.